Ramiro Benítez González

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Education

B.Sc. in Mathematics *Sep 2024 – Jun 2028 (Expected)*

University of Seville (US), Seville, Spain

Relevant coursework: Probability Theory (10.0, *Matrícula de Honor*) · Elements of Probability and Statistics (7.9) · Numerical Analysis I & II · Ordinary Differential Equations · Discrete Mathematics

Current year (2026–27): *First semester:* Statistical Inference · Complex Analysis · Algebraic Structures · Local Geometry of Curves and Surfaces · Advanced Differential Equations. *Second semester:* Mathematical Programming · Mathematical Modelling · Linear Models and Design of Experiments · Cryptography.

Academic Honors & Distinctions

"Matrícula de Honor" (Highest Distinction, max. 5% of students) in Probability Theory *2026*

Mathematical Olympiad – Participant, representing La Salle School (Córdoba) *2020*

Research Experience

Undergraduate Researcher ("Alumno Interno") *Oct 2026 – Present*

Department of Statistics and Operations Research (EIO), University of Seville

Project: Portfolio optimization. Mean-risk models and their LP/MILP formulations, risk measures, and estimation of expected returns and covariances from market data. [View Project]

Supervisors: Justo Puerto Albandoz and Alberto Torrejón Valenzuela

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Research Collaborator *Jun 2026 – Sep 2026*

Institute of Mathematics of the University of Seville (IMUS)

Project: Storage Location Assignment Problem (slotting) for an automated storage and picking system.

- MILP model for joint channel and replenishment-zone assignment (contiguity, three-level inventory, McCormick linearisation).
- Three-phase heuristic in Python for large instances: $O(1)$ cost preprocessing, greedy construction by Cube-per-Order Index and Iterated Local Search with ruin-and-recreate.
- Technical report in LaTeX and presentation of the algorithm to the research team.

Supervisors: Justo Puerto Albandoz and Alberto Torrejón Valenzuela

Research Interests

Operations Research & Optimization · Quantitative Finance · Probability & Stochastic Processes

Independent Study

Mansini et al., *Linear and Mixed Integer Programming for Portfolio Optimization* (Springer, 2015) and the **Gurobi Finance** models; reverse-engineered the Gurobi Finance dataset. [View Notebook]

Teaching & Other Experience

Private Tutor in Mathematics and Physics (self-employed): university entrance exam preparation.

Warehouse Logistics Assistant (Córdoba): inventory control and storage-space organisation.

Volunteer, Córdoba Food Bank and Adoratrices Association: collection and distribution campaigns.

Skills

- **Languages:** Spanish (Native), English (B1, *Cambridge B1 Preliminary*)
- **Programming: Python** (pandas, NumPy, SciPy, Matplotlib), **Gurobi** (LP, MILP, QP), **R**, C++ (basic)
- **Technical:** **LaTeX** (reports and Beamer), Git & GitHub, Pandoc

References

Alberto Torrejón Valenzuela – Lecturer, Department of Statistics and Operations Research, University of Seville. Research supervisor (IMUS, 2026; Alumno Interno, 2026–27). atorrejon@us.es

Full academic transcript available upon request.